

Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
Prof. Dr. Sofia C. Olhede

Practice Exam — Week 3: Backshift, Wold Decomposition, Causality & Invertibility

Duration: 60 minutes · No calculator · Answer on paper

Name: _____ Surname: _____

SCIPER (Student Card Number): _____

Exercise	Points
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2 .

Problem 1 (5 points)

(a) Define the *backshift operator* B and, for an ARMA(p, q) process written $\Phi(B)X_t = \Theta(B)\varepsilon_t$, write down the characteristic polynomials $\Phi(z)$ and $\Theta(z)$. State the *root criteria* (in terms of the unit circle) for (i) stationarity/causality and (ii) invertibility. (2 pts)

(b) Consider the AR(2) process

$$X_t = \frac{2}{3}X_{t-1} - \frac{1}{6}X_{t-2} + \varepsilon_t.$$

Write its AR polynomial $\Phi(z)$, find the roots, and decide whether the process is stationary. (2 pts)

(c) Is the process of part (b) invertible? Justify in one sentence (no computation needed). (1 pt)

Problem 2 (5 points)

For the ARMA(2, 1) process

$$(1 - 0.7B + 0.1B^2) X_t = (1 + 0.5B) \varepsilon_t,$$

(a) factor the AR polynomial $\Phi(z) = 1 - 0.7z + 0.1z^2$, locate its roots, and confirm that the process is causal. (2 pts)

(b) Using a *partial-fraction* expansion of the transfer function $G(z) = \Theta(z)/\Phi(z)$, find a closed form for the MA(∞) weights ψ_j in $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$, and report ψ_0, ψ_1, ψ_2 . (3 pts)

Problem 3 (5 points)

Let $\{X_t\}$ be the causal AR(2) process

$$X_t = \frac{5}{6}X_{t-1} - \frac{1}{6}X_{t-2} + \varepsilon_t.$$

(a) Show that the autocovariance sequence satisfies the homogeneous recurrence $\gamma_\tau = \frac{5}{6}\gamma_{\tau-1} - \frac{1}{6}\gamma_{\tau-2}$ for $\tau \geq 1$, and obtain the boundary relation at $\tau = 0$ involving σ^2 . (2 pts)

(b) Solve for γ_0, γ_1 and hence give the closed-form expression of γ_τ for all $\tau \geq 0$. (3 pts)

Hint: the right-hand side of the model equation only contributes through $\mathbb{E}[\varepsilon_t X_t] = \sigma^2$ (causality).

Problem 4 (5 points)

(Revision of Week 2 + this week.) Let $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$ be an MA(1) with $\theta = \frac{1}{2}$.

(a) Compute the autocovariance sequence γ_τ for all τ and the autocorrelation ρ_1 . (1.5 pts)

(b) Prove that *any* MA(1) of the form $\varepsilon_t - \theta\varepsilon_{t-1}$ satisfies $|\rho_1| \leq \frac{1}{2}$, and state qualitatively how the PACF of an MA(1) behaves (cut off or tail off?). (1.5 pts)

(c) Is $\{X_t\}$ invertible? Then exhibit a *different* MA(1) process $Y_t = \eta_t - \vartheta\eta_{t-1}$ (with an appropriately chosen white-noise variance) that has *exactly the same* autocovariance sequence as $\{X_t\}$, and explain which of the two is the invertible representation. (2 pts)

